



Mathematics
Pathway 2
End of Year Test

Baldivis
Secondary College

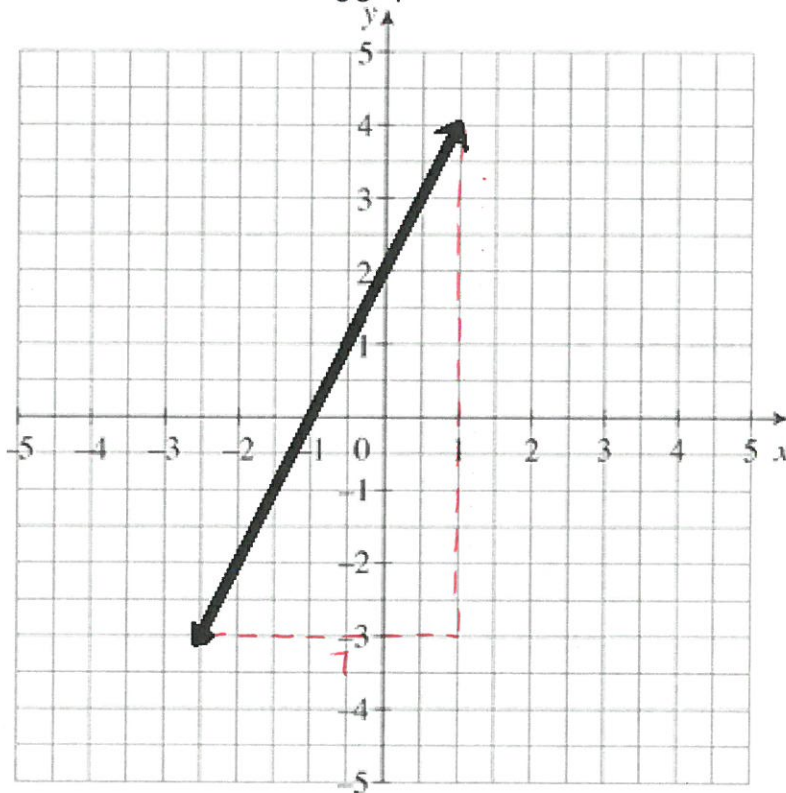
Name: Solutions. Class: _____ Score: _____
Resources Allowed: Calculator, 1 page of notes (A4 two-sided), Formula sheet

Time: 55 mins

Marks: 40

1. [3, 2, 2 = 7 marks]

In reference to the following graph



a) State the;
i) gradient

2 ✓

ii) y-intercept

2 ✓

iii) x-intercept

-1 ✓

b) Write the equation for the line

$$y = 2x + 2 \quad \checkmark \checkmark$$

c) What is the length of the line?

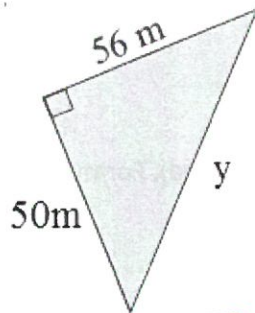
$$7^2 + 14^2 = 245 \quad \checkmark$$

$$\sqrt{245} = 15.65 \quad \checkmark (2dp)$$

2. [2, 2 = 4 marks]

Determine the value of the missing side using Pythagoras

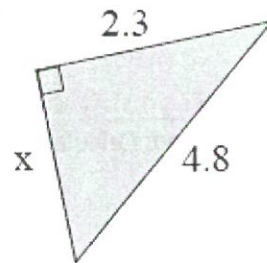
a)



$$y = \sqrt{56^2 + 50^2} \quad \checkmark$$

$$= 75.07\text{m} \quad (2\text{dp}) \quad \checkmark$$

b)



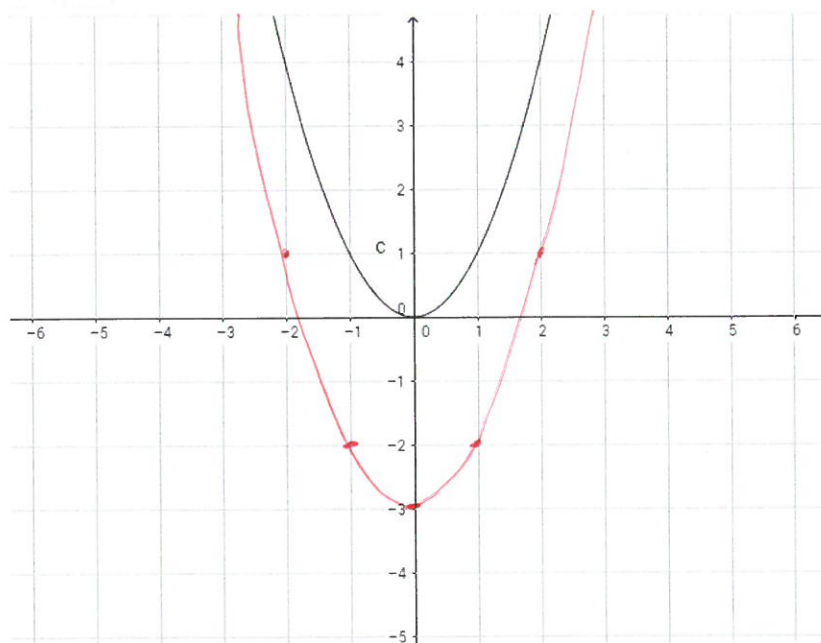
$$x = \sqrt{4.8^2 - 2.3^2} \quad \checkmark$$

$$= 4.21 \quad (2\text{dp}) \quad \checkmark$$

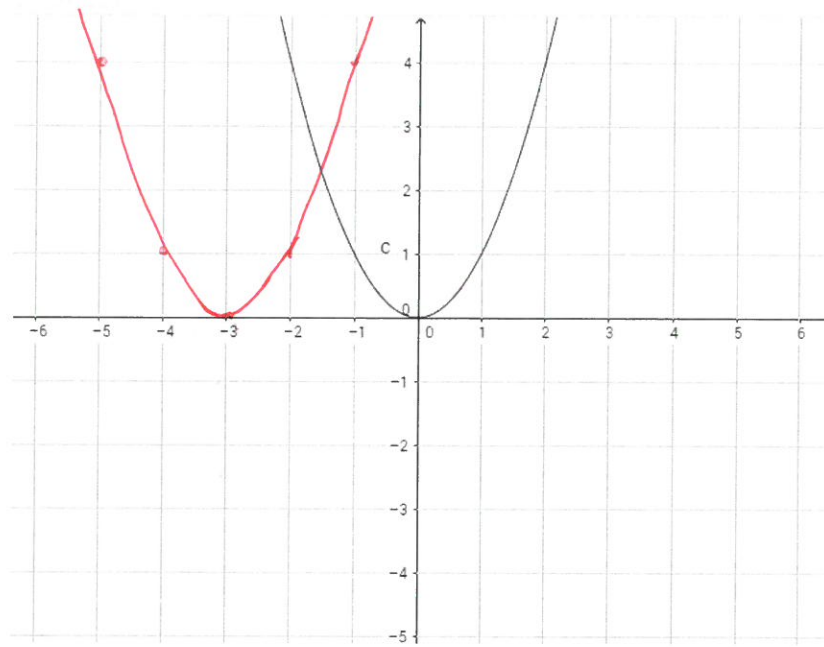
3. [2, 2, 2 = 6 marks]

On the graphs $y = x^2$ below:

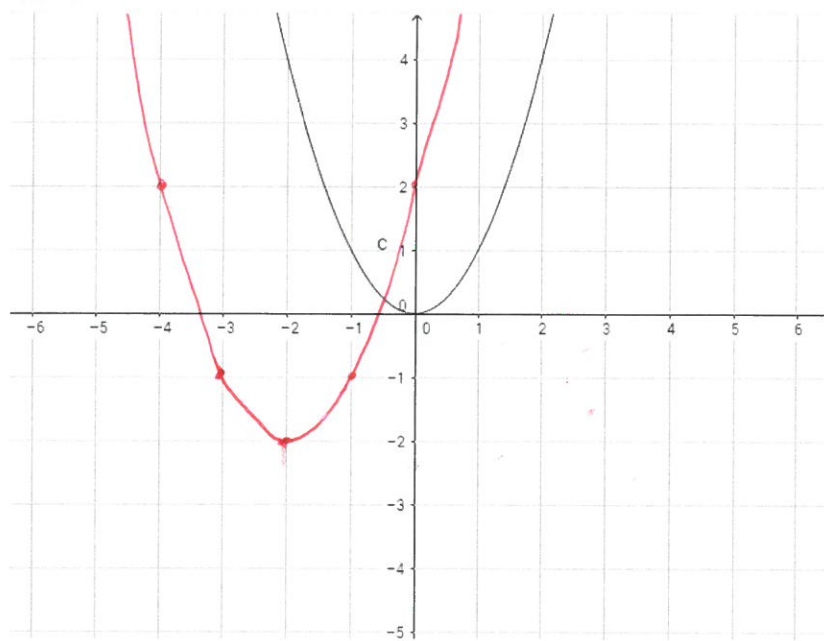
a) Plot the graph $y = x^2 - 3$



b) $y = (x+3)^2$



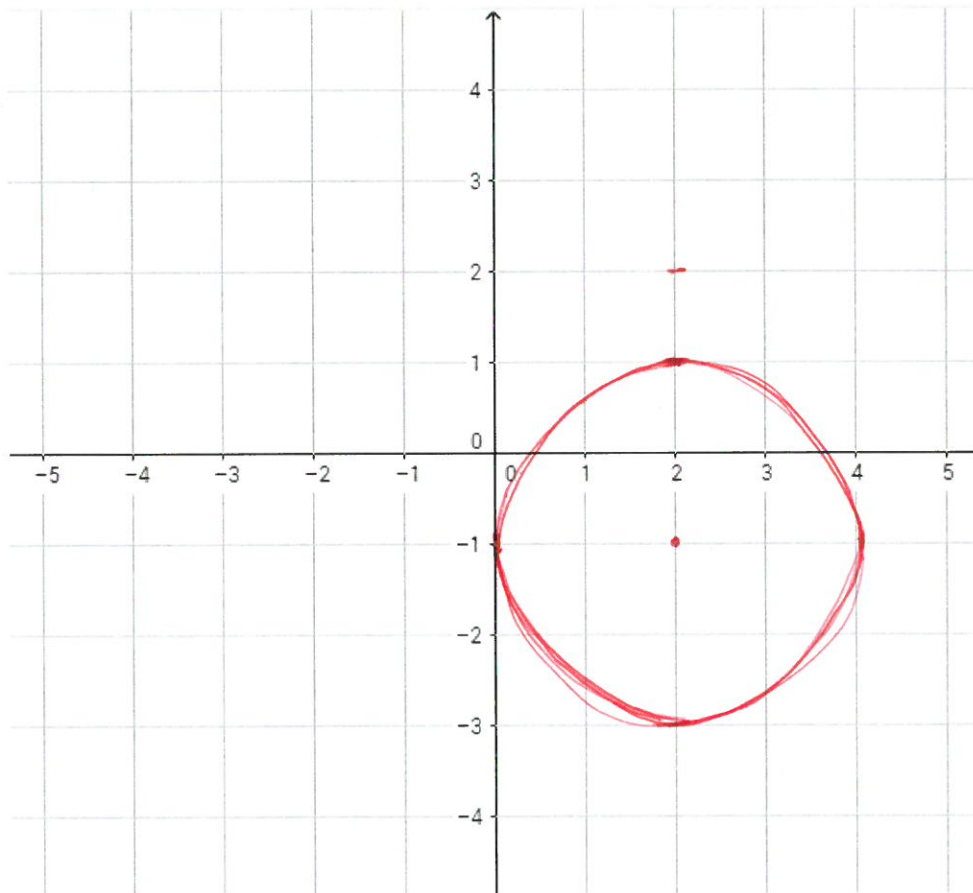
c) $y = (x+2)^2 - 2$



4. [2, 2 = 4 marks]

a) On the axes below, graph the circle $(x - 2)^2 + (y + 1)^2 = 4$

$(2, -1)$ ✓ $r = 2$ ✓



b) What is the equation of a circle with a centre at $(0, 3)$ with a radius of 2?

$$x^2 + (y - 3)^2 = 4$$

✓ ✓

5. (3, 3, 1 = 7 marks)

Use the following data set to answer these questions:

✓ 6 26 28 28 30 32 32 33 33 35 39

6	28	26	32	30	32	28	35	33	33	39
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a) Find the median

32 ✓✓

b) Find the mean

$$\frac{322}{11} = 29.27 \quad (2dp) \quad \checkmark \checkmark \checkmark$$

c) Find the mode

There is no mode. or multimodal.
✓

6. (3 Marks)

What is the effect on the mean, median and mode of removing the outlier from the above data set?

✓ mode - no change.

✓ mean - $\frac{316}{10} = 31.6$ mean increases.

✓ median - 32 no change.

7. (5, 4 = 9)

↓

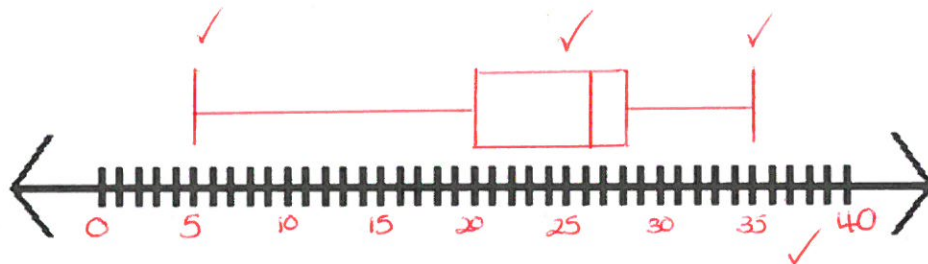
5	15	20	21	25	27	27	28	30	35
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a) Use the above data set to find the five number summary to construct a box and whisker plot.

Min	Q_1	Median	Q_3	Max
5	20	26	28	35

✓ ✓ ✓ ✓ ✓

b) Use your values from a) to construct a box and whisker plot in the space provided below:



End of Test